

☒ EASY

☐ ALGEBRA

Algebra

10 questions · ~15 min

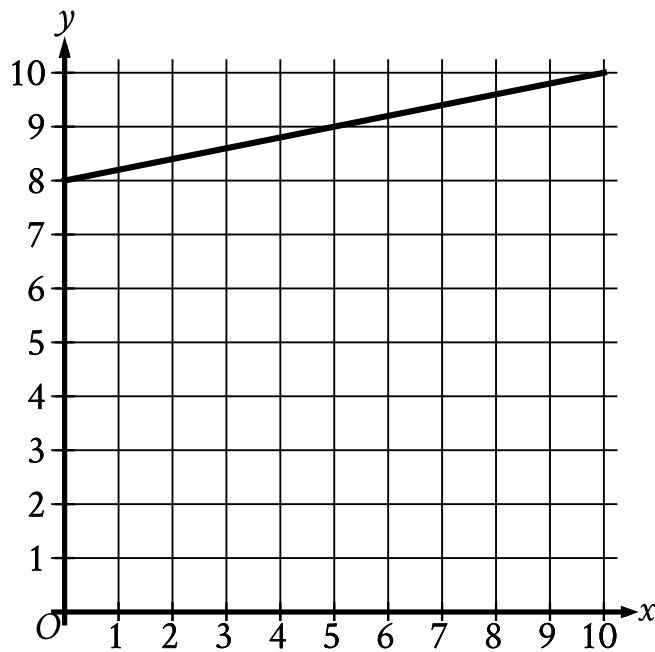
13

Q1

ALGEBRA

Ty set a goal to walk at least 24 kilometers every day to prepare for a multiday hike. On a certain day, Ty plans to walk at an average speed of 4 kilometers per hour. What is the minimum number of hours Ty must walk on that day to fulfill the daily goal?

- A. 4
- B. 6
- C. 20
- D. 24



- The line slants gradually up from left to right.
- The line passes through the following points:
 - (0 comma 8)
 - (5 comma 9)
 - (10 comma 10)

What is the y -intercept of the line graphed?

- A. 0, - 8
- B. 0, $-\frac{1}{8}$
- C. 0,0
- D. 0,8

Q3

GRID-IN

ALGEBRA

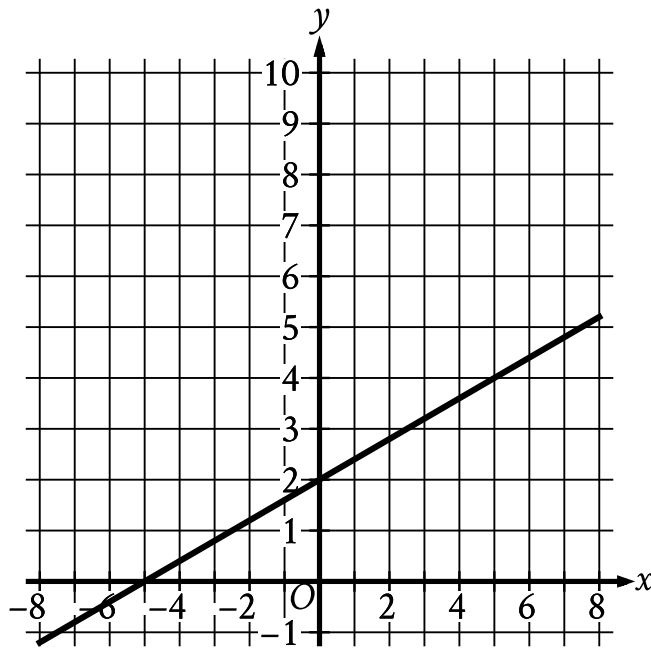
$$13x = 112 - x$$

What value of x is the solution to the given equation?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- The line slants gradually up from left to right.
- The line passes through the following points:
 - $(-5, 0)$
 - $(0, 2)$

The graph of the linear function f is shown. What is the y-intercept of the graph of $y = fx$?

- A. $-5, 0$
- B. $2, 0$
- C. $0, 2$
- D. $0, -5$

Q5**ALGEBRA**

$$x + y = 18$$

$$5y = x$$

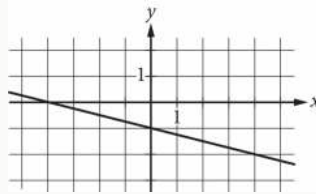
What is the solution x, y to the given system of equations?

- A. 15, 3
- B. 16, 2
- C. 17, 1
- D. 18, 0

Q6**ALGEBRA**

For a literature class, a student's essay needs to have no more than 5,000 words. If a student's essay already has 3,026 words, what is the maximum number of additional words the student can include in the essay?

- A. 1,974
- B. 3,026
- C. 5,000
- D. 8,026



Which of the following is an equation of the graph shown in the xy -plane above?

A. $y = -\frac{1}{4}x - 1$

B. $y = -x - 4$

C. $y = -x - \frac{1}{4}$

D. $y = -4x - 1$

Q8

GRID-IN ALGEBRA

If $2 + x = 60$, what is the value of $16 + 8x$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

ALGEBRA

Sean rents a tent at a cost of \$ 11 per day plus a onetime insurance fee of \$ 10. Which equation represents the total cost c , in dollars, to rent the tent with insurance for d days?

A. $c = 11d + 10$

B. $c = 10d + 11$

C. $c = 11d + 10$

D. $c = 10d + 11$

Q10

ALGEBRA

$$4x - 3y = 5$$

$$x = 8$$

What is the solution x, y to the given system of equations?

A. 8, 9

B. 8, -24

C. 8, -9

D. 8, 24